Exercises 7.1 3, 5, 7, 9, 11, 13, 15, 17, 19, 23, 27, 29, 33, 35, 37, 41, 47, 49, 51, 53, 61, 63, 65.

1–4 Evaluate the integral using integration by parts with the indicated choices of u and dv.

3. $\int x \cos 4x\, dx; \quad u = x, \; dv = \cos 4x\, dx$

$du = dx \qquad v = \frac{\sin 4x}{4}$

$= x \cdot \frac{\sin 4x}{4} - \int \frac{\sin 4x}{4}\, dx$

$= \frac{x \sin 4x}{4} + \frac{1}{16} \cos 4x$

5–42 Evaluate the integral.

5. $\int_0^4 te^{2t}\, dt$ LIATE

let $u = e^{2t}$; $du = 2e^{2t}\, dt$

$dv = t\, dt$; $v = \frac{t^2}{2}$ ✗

$= \frac{1}{2} e^{2t} \cdot t^2 - \int \frac{t^2}{2} \cdot 2e^{2t}\, dt$

$= \frac{1}{2} e^{2t} \cdot t^2 - \int t^2 \cdot e^{2t}\, dt$

let $u = t^2$; $du = 2t\, dt$

$dv = e^{2t} dt$; $v = \frac{e^{2t}}{2}$

$= \frac{1}{2} e^{2t} \cdot t^2 - \left(\frac{t^2 \cdot e^{2t}}{2} - \int \frac{e^{2t}}{2} \cdot 2t\, dt \right)$

$= \frac{1}{2} e^{2t} \cdot t^2 - \frac{t^2 \cdot e^{2t}}{2} + \int e^{2t} t\, dt$

let $u = t$; $du = 1\, dt$

$dv = e^{2t} dt$; $v = \frac{e^{2t}}{2}$

$= \cdots + \left(\frac{t \cdot e^{2t}}{2} - \int \frac{e^{2t}}{2} dt \right) = \cdots + \frac{te^{2t}}{2} - \frac{e^{2t}}{4} + C$

let $u = t$; $du = dt$

$dv = e^{2t} dt$; $v = \frac{e^{2t}}{2}$

$= \frac{t \cdot e^{2t}}{2} - \int \frac{e^{2t}}{2} dt$

$= \frac{t \cdot e^{2t}}{2} - \frac{e^{2t}}{4} + C$

7. $\int x \sin 10x \, dx$ LIATE

Let $u = x$; $du = dx$

$dv = \sin 10x \, dx$; $v = \frac{-\cos 10x}{10}$

$= \frac{-x \cdot \cos 10x}{10} - \int \frac{-\cos 10x}{10} dx$ $+ \frac{1}{10} \int \cos 10x \, dx$

$= -\frac{x \cos 10x}{10} + \frac{1}{10} \int \cos 10x \, dx$

$= -\frac{x \cos 10x}{10} + \frac{1}{100} \sin 10x + C$

11. $\int (x^2 + 2x) \cos x \, dx$

let $u = x^2 + 2x$; $du = 2x + 2 \, dx$

$dv = \cos x \, dx$; $v = \sin x$

$= (x^2 + 2x) \sin x - \int \sin x \, (2x + 2) \, dx + C$

let $u = 2x + 2$; $du = 2 \, dx$

$dv = \sin x \, dx$; $v = -\cos x$

$= (x^2 + 2x) \sin x - \left(-(2x + 2) \cos x - \int -\cos x \cdot 2 \, dx \right) + C$

$= (x^2 + 2x) \sin x + (2x + 2) \cos x - 2 \int \cos x \, dx + C$

$= \ldots + \ldots - 2 \sin x + C$

13. $\int \cos^{-1}x\,dx$

let $u = \cos^{-1}x$; $du = -\frac{1}{\sqrt{1-x^2}}dx$

$dv = dx$; $v = x$

$= x\cos^{-1}x - \int x\left(-\frac{1}{\sqrt{1-x^2}}dx\right)$

$= x\cos^{-1}x + \int x\cdot\frac{1}{\sqrt{1-x^2}}dx$

substitution

let $u = x$; $du = dx$

$dv = \frac{1}{\sqrt{1-x^2}}dx$; $v = \cos^{-1}x$

$= x\cos^{-1}x + x\cos^{-1}x - \int \cos^{-1}x\,dx$

let $u = 1-x^2$; $\frac{du}{dx} = -2x \Rightarrow dx = \frac{du}{-2x}$

$= x\cos^{-1}x + \int \frac{x}{\sqrt{u}}\cdot\frac{du}{-2x}$

$= x\cos^{-1}x - \frac{1}{2}\int \frac{1}{\sqrt{u}}du$

$= x\cos^{-1}x - u^{\frac{1}{2}} + C$

$= x\cos^{-1}x - \sqrt{1-x^2} + C$

15. $\int t^4 \ln t\,dt$ LIATE

let $u = \ln t$; $du = \frac{1}{t}dt$

$dv = t^4\,dt$; $v = \frac{t^5}{5}$

$= \frac{1}{5}\cdot t^5\ln t - \frac{1}{5}\int t^5\cdot\frac{1}{t}dt$

$= \frac{1}{5}t^5\ln t - \frac{1}{25}t^5 + C$

17. $\int t \csc^2 t \, dt$ LIATE

let $u = t$; $du = dt$

$dv = \csc^2 t \, dt$; $v = -\cot x$

$= -t\cot x - \int -\cot x \, dt$

$= -t\cot x + \int \cot x \, dx$

let $u = \cot x$; $du = -\csc^2 x$

$dv = dx$; $v = x$

$= -t\cot x + x\cot x - \int x -$ $= -t\cot t + \int \frac{\cos t}{\sin t} dt$

let $u = \sin t$; $du = \cos t \, dt$

$dt = \frac{du}{\cos t}$

$= -t\cos t + \int \frac{\cos t}{u} \cdot \frac{dt}{\cos t}$

$= -t\cos t + \ln|\sin t| + C$

19. $\int (\ln x)^2 \, dx$

Let $u = (\ln x)^2$; $du = 2\ln x \cdot (\frac{1}{x}) dx$

$dv = dx$; $v = x$

$= x(\ln x)^2 - \int x \cdot 2\ln x \cdot (\frac{1}{x}) dx$

$= x\ln^2 x - 2\int \ln x \, dx$

Let $u = \ln x$; $du = \frac{1}{x} dx$

$dv = dx$, $v = x$

$= x\ln^2 x - 2(x\ln x - \int x \cdot \frac{1}{x} dx)$

$= x\ln^2 x - 2(x\ln x - x + C_1)$

$= x\ln^2 x - 2x\ln x + 2x + C$, where $C = -2C_1$

这个点没看懂.

23. $\int e^{2\theta} \sin 3\theta \, d\theta$ (T) LIATE

let $u = \sin 3\theta$, $du = 3\cos 3\theta \, d\theta$

$dv = e^{2\theta} d\theta$, $v = \frac{e^{2\theta}}{2}$

$= \frac{1}{2} e^{2\theta} \cdot \sin 3\theta - \frac{3}{2} \int e^{2\theta} \cdot \cos 3\theta \, d\theta$

let $u = \cos 3\theta$, $du = -3\sin 3\theta \, d\theta$

$dv = e^{2\theta} d\theta$, $v = \frac{e^{2\theta}}{2}$

$= \cdots - \frac{3}{2}\left(\frac{e^{2\theta}}{2} \cos 3\theta - \int \frac{e^{2\theta}}{2} (-3\sin 3\theta) \, d\theta \right)$

与原式重复.

$\int e^{2\theta} \sin 3\theta \, d\theta = \frac{1}{2} e^{2\theta} \cdot \sin 3\theta - \frac{3}{4} e^{2\theta} \cos 3\theta - \frac{9}{4} \int e^{2\theta} \sin 3\theta \, d\theta$

$\frac{13}{4} \int e^{2\theta} \sin 3\theta \, d\theta = \frac{1}{2} e^{2\theta} \sin 3\theta - \frac{3}{4} e^{2\theta} \cos 3\theta + C_1$

$\int e^{2\theta} \sin 3\theta \, d\theta = \frac{2}{13} e^{2\theta} \sin 3\theta - \frac{3}{13} e^{2\theta} \cos 3\theta + C$

where $C = \frac{4}{13} C_1$

27. $\int (1 + x^2)\, e^{3x}\, dx$

let $u = 1+x^2$; $du = 2x\,dx$

$dv = e^{3x}dx$; $v = \frac{e^{3x}}{3}$

$= \frac{1}{3} e^{3x}(1+x^2) - \frac{2}{3}\int e^{3x} \cdot x\,dx$

let $u = x$, $du = dx$

$dv = e^{3x}dx$, $v = \frac{e^{3x}}{3}$

$= \frac{1}{3} e^{3x}(1+x^2) - \frac{2}{3}\left(\frac{1}{3} e^{3x} \cdot x - \frac{1}{3}\int e^{3x}\,dx\right)$

~~$= \frac{1}{3} e^{3x}(1+x+x^2) + \frac{1}{9} e^{3x} + C$~~ x

$= \frac{1}{3} e^{3x} + \frac{1}{3} e^{3x} \cdot x^2 - \frac{2}{9} e^{3x} \cdot x + \frac{2}{27} e^{3x} - \frac{2}{3} C_1$

$= \frac{11}{27} e^{3x} - \frac{2}{9} x\, e^{3x} + \frac{1}{3} x^2 e^{3x} + C$, where $C = -\frac{2}{3} C_1$

29. $\int_0^1 x\,3^x\,dx$

6

let $u = x$; $du = dx$

$dv = 3^x dx$, $v = \frac{3^x}{\ln 3}$

$= \left[\frac{x \cdot 3^x}{\ln 3}\right]_0^1 - \int_0^1 \frac{3^x}{\ln 3}\,dx$

$= \frac{x \cdot 3^x}{\ln 3} - \frac{3^x}{\ln^2 3} + C \left[\frac{3^x}{\ln^2 3}\right]_0^1$

$= \frac{1}{\ln 3}[3 - 0] - \frac{1}{\ln^2 3}[3 - 1]$

$= \frac{3}{\ln 3} - \frac{2}{(\ln 3)^2}$

33. $\int_1^5 \frac{\ln R}{R^2}\, dR$

let $u = \ln R$; $du = \frac{1}{R}\, dR$

$dv = \frac{1}{R^2}\, dR$; $v = -R^{-1}$

$= \left[-\frac{\ln R}{R}\right]_1^5 - \int -\frac{1}{R} \cdot \frac{1}{R}\, dR$

$= -\frac{\ln R}{R} + \int \frac{1}{R^2}\, dR$

$= -\left[\frac{\ln R}{R}\right]_1^5 - \left[\frac{1}{R}\right]_1^5$

$= -\frac{\ln 5}{5} - \left(\frac{1}{5} - 1\right) = \frac{4}{5} - \frac{\ln 5}{5}$



$\int_0^{\pi} \cos 2x\, dx$

$\sin 2x$

$\frac{}{2}$

35. $\int_0^{\pi} x \sin x \cos x\, dx$ LIATE

$= \int_0^{\pi} x \frac{\sin 2x}{2}\, dx$ $\sin 2x$

let $u = x$; $du = dx$

$dv = \sin 2x\, dx$; $v = -\frac{\cos 2x}{2}$

$= \frac{1}{2}\left(-\frac{x \cos 2x}{2} - \int_0^{\pi} \frac{-\cos 2x}{2}\, dx\right)$

$= -\frac{1}{4}\left[x \cos 2x\right]_0^{\pi} + \frac{1}{8}\left[\sin 2x\right]_0^{\pi}$

$= -\frac{1}{4}(2\pi - 0) + \frac{1}{8} \times 0 = -\frac{\pi}{2} \times -\frac{\pi}{4}$

37. $\int_1^5 \frac{M}{e^M}\, dM$

Let $u = M$; $du = dM$

$dv = \frac{1}{e^M} dM$; $v = -e^{-M}$

$= [M \cdot e^{-M}]_1^5 - \int_1^5 e^{-M}\, dM$

$= -(5 \cdot e^{-5} - e^{-1}) - [e^{-M}]_1^5$

$= 0.334 - (e^{-5} - e^{-1})$

$= 0.334 + 0.361 = 0.695$

47. $\int x \ln(1+x)\,dx$

$u = \ln(1+x)$; $du = \frac{1}{1+x}\,dx$

$dx = (1+x)\,du$

let $y = 1+x$; $dy = dx$ ($x = y-1$)

$\int (y-1)\cdot \ln y\,dy$

let $u = \ln y$; $du = \frac{1}{y}\,dy$

$dv = (y-1)\,dy$, $v = \frac{y^2}{2} - y$

$= (\frac{y^2}{2} - y)\ln y - \int (\frac{y^2}{2} - y)\frac{1}{y}\,dy$

$= \frac{y^2}{2}\ln y - y\ln y - \int \frac{y}{2} - 1\,dy$

$= \frac{y^2}{2}\ln y - y\ln y - \frac{y^2}{4} - y + C$

$y = 1+x$

$\Rightarrow = \frac{(1+x)^2}{2}\ln(1+x) - (1+x)\ln(1+x) - \frac{(1+x)^2}{4} - (1+x) + C$

49. $\int xe^{-2x}\,dx$

Let $u = x$; $du = dx$

$dv = e^{-2x}dx$; $v = \frac{e^{-2x}}{-2}$

$= -\frac{1}{2}(x \cdot e^{-2x}) + \frac{1}{2}\int e^{-2x}dx$

$= -\frac{x \cdot e^{-2x}}{2} - \frac{1}{4} \cdot e^{-2x} + C$

51. $\int x^3\sqrt{1+x^2}\,dx$

substitution

$= \int x^2 \cdot x\sqrt{1+x^2}\,dx$

Let $u = 1+x^2$; $\frac{du}{dx} = 2x$; $dx = \frac{du}{2x}$

$x^2 = u-1$

$= \int x^2 \cdot x\sqrt{u} \cdot \frac{1}{2x}\,du$

$= \frac{1}{2}\int x^2 \cdot \sqrt{u}\,du$

$= \frac{1}{2}\int (u-1)\sqrt{u}\,du$

$= \frac{1}{2}\int u^{\frac{3}{2}} - u^{\frac{1}{2}}\,du$

$= \frac{1}{2} \times \frac{2}{7}u^{\frac{7}{2}} - \frac{1}{2} \times \frac{2}{3} \times u^{\frac{3}{2}} + C$

$= \frac{1}{7}u^{\frac{7}{2}} - \frac{1}{3}u^{\frac{3}{2}} + C$

53. (a) Use the reduction formula in Example 6 to show that

$$\int \sin^2 x\,dx = \frac{x}{2} - \frac{\sin 2x}{4} + C$$

(b) Use part (a) and the reduction formula to evaluate $\int \sin^4 x\,dx$.

a) $\int \sin^2 x\,dx = \int \frac{1-\cos 2x}{2}\,dx$

$= \int \frac{1}{2}\,dx - \frac{1}{2}\int \cos 2x\,dx$

$= \frac{x}{2} - \frac{1}{2}\times\frac{1}{2}\sin 2x + C$

$= \frac{x}{2} - \frac{\sin 2x}{4} + C$

b) $\int \sin^4 x\,dx = \int (\sin^2 x)^2\,dx$